AI Specialist

AI Specialist Project On:
CADGAURD-AI

INSTITUTE OF AERONAUTICAL ENGINEERING

(Autonomous)

Dundigal, Hyderabad – 500 043, Telangana

Department of CSE

Bachelor of Technology

in

CSE

-A TEAM PROJECT BY

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Certificate

This is to certify that it is a bonafied record of practical work done by Mr. / Ms.

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of_____class_____ branch in

the _____laboratory during the academic

year _____ under our supervision.

Head of the department **Lecturer – in charge**

Signature of External Examiner **Signature of Internal Examiner**

DECLARATION

I certify that

a. The work contained in this report is original and has been done by me under the guidance of my supervisor (s).
b. The work has not been submitted to any other Institute for any degree or diploma.
c. I have followed the guidelines provided by the Institute for preparing the report.
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ABSTRACT

CADGaurd is an AI-augmented CAD quality platform that streamlines the process of analyzing and validating engineering designs. It combines deterministic geometry checks, lightweight digital twin–based stress estimation, and generative AI assistance into a single workflow. This integration enables engineers to quickly identify design flaws, assess structural risks, and receive clear, actionable feedback without extensive manual analysis, thereby improving both efficiency and design reliability.

The platform uses a React and TypeScript frontend to provide a structured and interactive user experience. It supports secure project management, 3D model visualization, validation review, and simulation insights within a unified interface. Additionally, conversational AI features act as a copilot, helping users understand detected issues and guiding them with optimization suggestions in a simplified and accessible manner.

On the backend, Supabase manages authentication, data storage, and real-time synchronization. It maintains persistent records for projects, issues, simulations, and reports, ensuring data consistency and enabling collaborative workflows. The computational processing is handled by a FastAPI backend, which ingests CAD meshes, extracts geometric features, performs rule-based manufacturability checks, and estimates stress distribution to detect weak regions and assign risk levels.

An AI layer powered by Groq enhances the system by converting technical outputs into understandable explanations, summaries, and recommendations. It also supports chatbot interactions and auto-fix suggestions, reducing the effort required for interpretation and decision-making. Overall, CADGaurd's modular and production-ready architecture allows engineers to move from raw CAD input to explainable quality assessment efficiently, making it a valuable tool for modern engineering design workflows.

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CHAPTER 1
INTRODUCTION

1.1.1 About Project Objectives

CADGaurd is an end-to-end CAD validation platform that enables engineers to upload models, perform automated geometry and manufacturability checks, and review results within a unified workflow. It detects risks early using deterministic rules such as wall thickness, watertightness, and structural analysis, while providing fast, material-aware stress estimation. The system integrates

1.2 Requirements

Requirements behind CADGaurd is twofold:

- **Functional:** Provide users with a unified, secure platform to upload CAD models, perform automated validation and stress abalysis, and receive AI-driven insights and recommendations for improving design quality.
- **Technical (DevOps):** Demonstrate how DevOps principles can improve the development, deployment, and maintenance of modern software applications.

1.3 Scope

CADGaurd aims to provide:

- CADGaurd aims to provide a unified platform for uploading and managing CAD models.
- CADGaurd aims to provide automated validation of geometry and manufacturability.
- CADGaurd aims to provide fast stress analysis to identify weak regions and risks.
- CADGaurd aims to provide AI-driven insights and actionable design recommendations.
- CADGaurd aims to provide secure, real-time access with dashboards and collaborative features.

DevOps Scope:

- GitHub for SCM
- CI/CD with Jenkins
- Docker for packaging
- Kubernetes for deployment
- Monitoring with Prometheus and Grafana

CHAPTER 2

METHODOLGY

DevOps is a cultural and technical movement that aims to unify software development (Dev) and IT operations (Ops). The primary goal of DevOps is to shorten the software development life cycle while delivering features, fixes, and updates frequently, reliably, and in close alignment with business objectives.

The methodology used in this project is rooted in DevOps practices, especially those related to automation, continuous feedback, and collaborative development. The CADGaurd AI built with GitHub Actions is a real-world implementation of DevOps ideology—focusing on **continuous integration**, **continuous delivery**, and **continuous deployment**.

This project follows several core DevOps principles in its methodology:

1. **Automation**: Manual tasks such as testing, building, and deploying are automated through GitHub Actions workflows.

2. **Collaboration**: All code, including automation scripts and configurations, is version-controlled and accessible to team members via GitHub.

3. **Continuous Integration (CI)**: Every code commit triggers an automated process that builds the application and runs tests.

4. **Continuous Deployment (CD)**: When CI is successful, the new version is automatically deployed to a server or cloud environment.

5. **Monitoring and Feedback**: Logs and status checks from each step provide immediate feedback to developers.

Phases of the Methodology

The CADGaurd AI implementation in this project can be broken down into several interconnected phases, all aligned with DevOps workflow:

1. Code Development and Version Control

The process begins with developers writing application code using a modular and testable architecture. Code is regularly committed to a GitHub repository using Git. Branching strategies (e.g., feature, develop, and main) may be used to maintain clean integration workflows.

Tools involved:

- Git (for version control)
- GitHub (for hosting the repository and managing collaboration)

2. Workflow Configuration with GitHub Actions

The .github/workflows/ci.yml file is created in the repository to define the CADGaurd AI using **YAML syntax**. This file contains instructions for:

- The triggering event (e.g., push, pull request)
- The jobs to run (e.g., test, build, deploy)
- The sequence of steps in each job

3. Continuous Integration

Whenever a developer pushes code to the repository, GitHub Actions automatically triggers the **CI workflow**. This workflow typically includes:

- **Dependency Installation**: Installing project dependencies (e.g., via pip, npm, or maven)
- **Linting and Static Code Analysis**: Ensuring code quality and formatting
- **Unit Testing**: Running automated tests to validate new changes

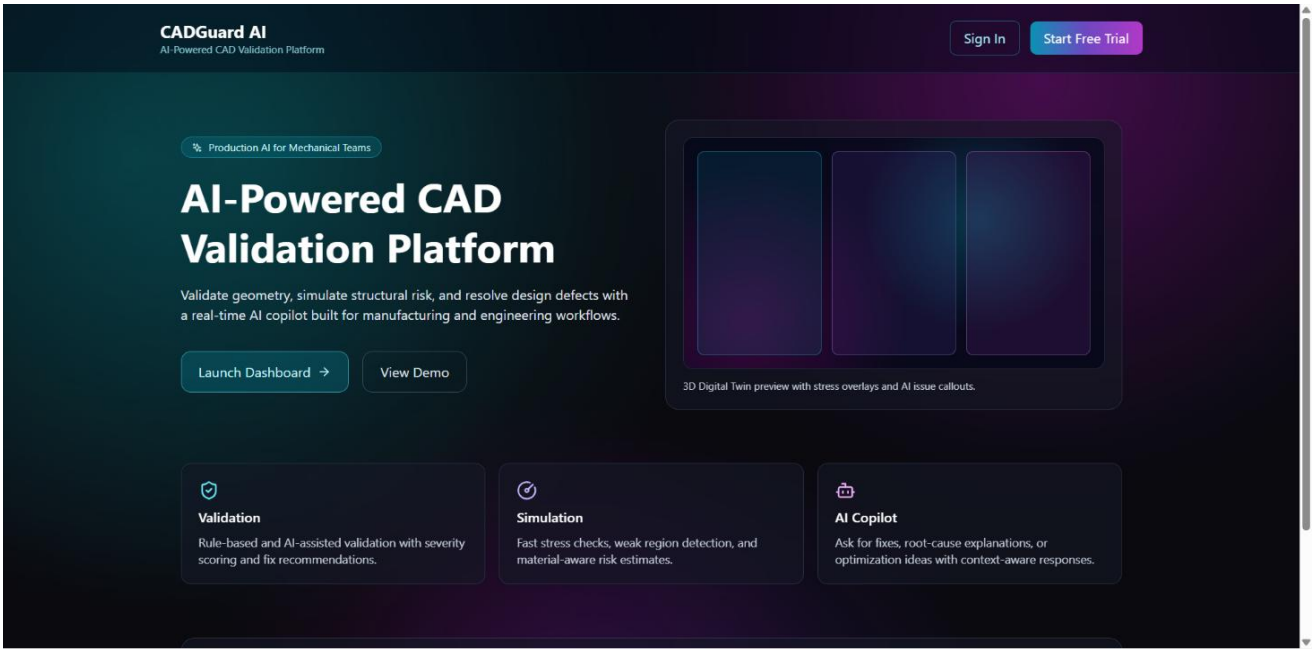
4. Continuous Delivery and Deployment

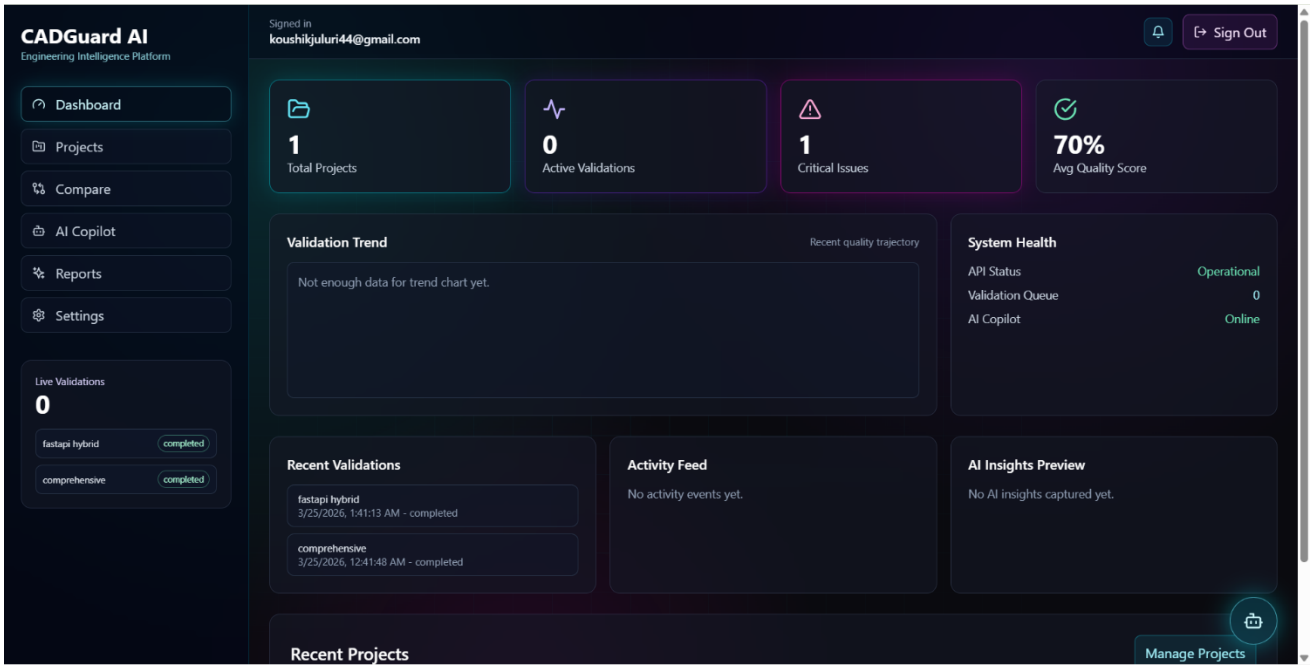
Once the CI phase is complete and successful, where the application is built (e.g., compiled or containerized) and deployed to a designated environment.
Depending on the project scope, the deployment target can be:

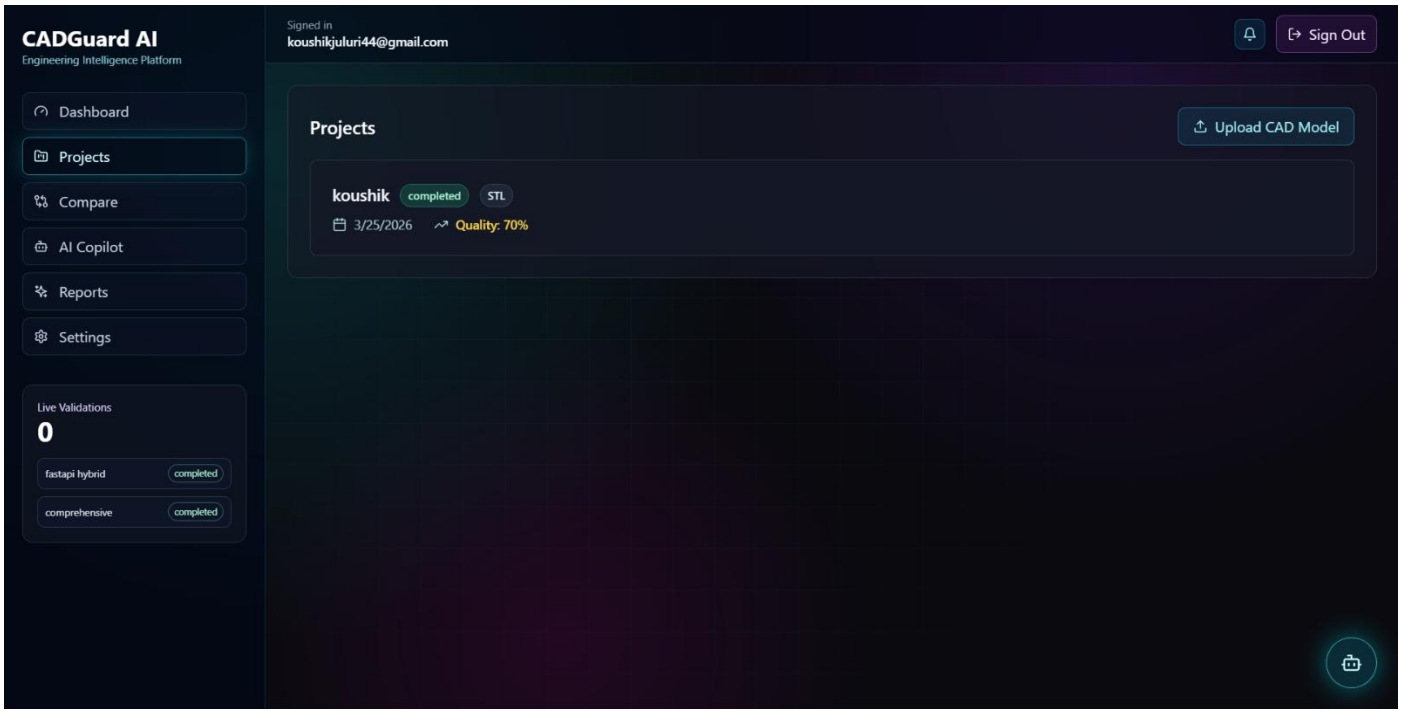
- A static hosting service (e.g., GitHub Pages)
- A cloud-based PaaS (e.g., Heroku, AWS Elastic Beanstalk)
- A container registry and orchestration platform (e.g., Docker Hub + Kubernetes)

CHAPTER 3
RESULTS AND DISCUSSION

Execution:



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Dashboard
Projects
Compare
AI Copilot
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Projects
Upload CAD Model
koushik
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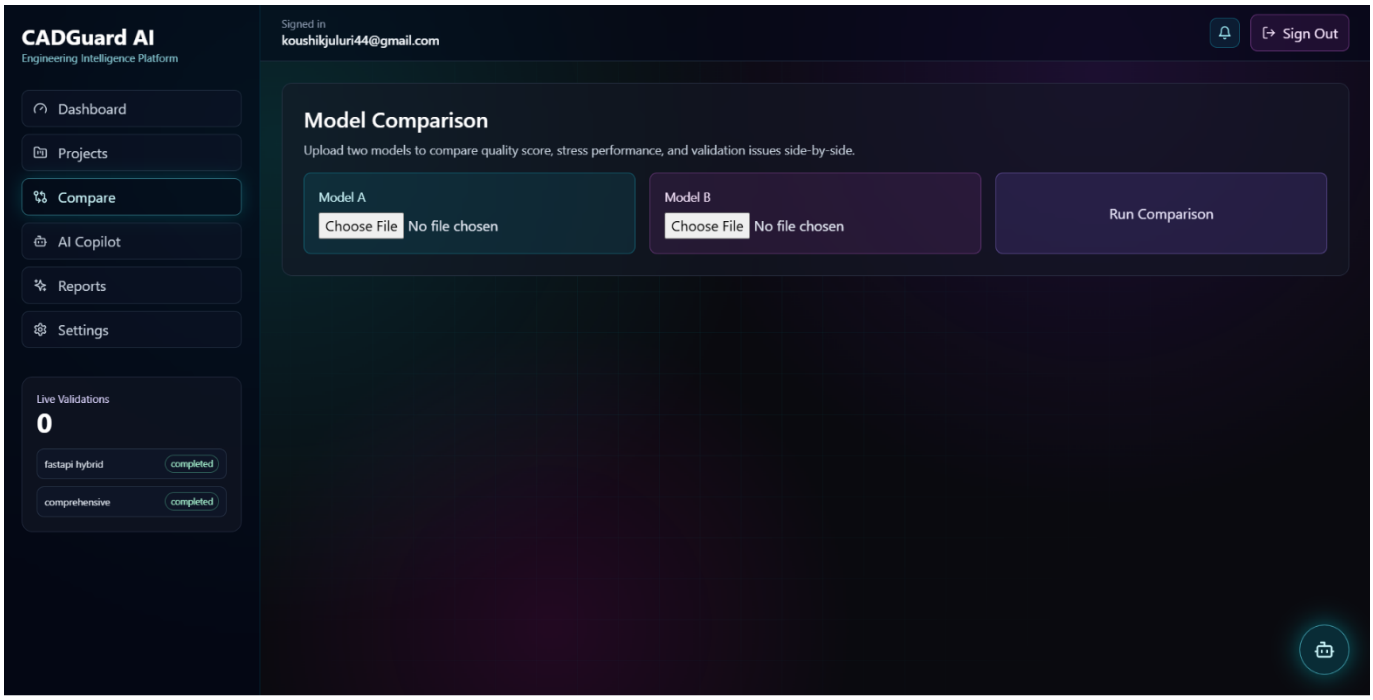
STL
3/25/2026
Quality: 70%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Live Validations
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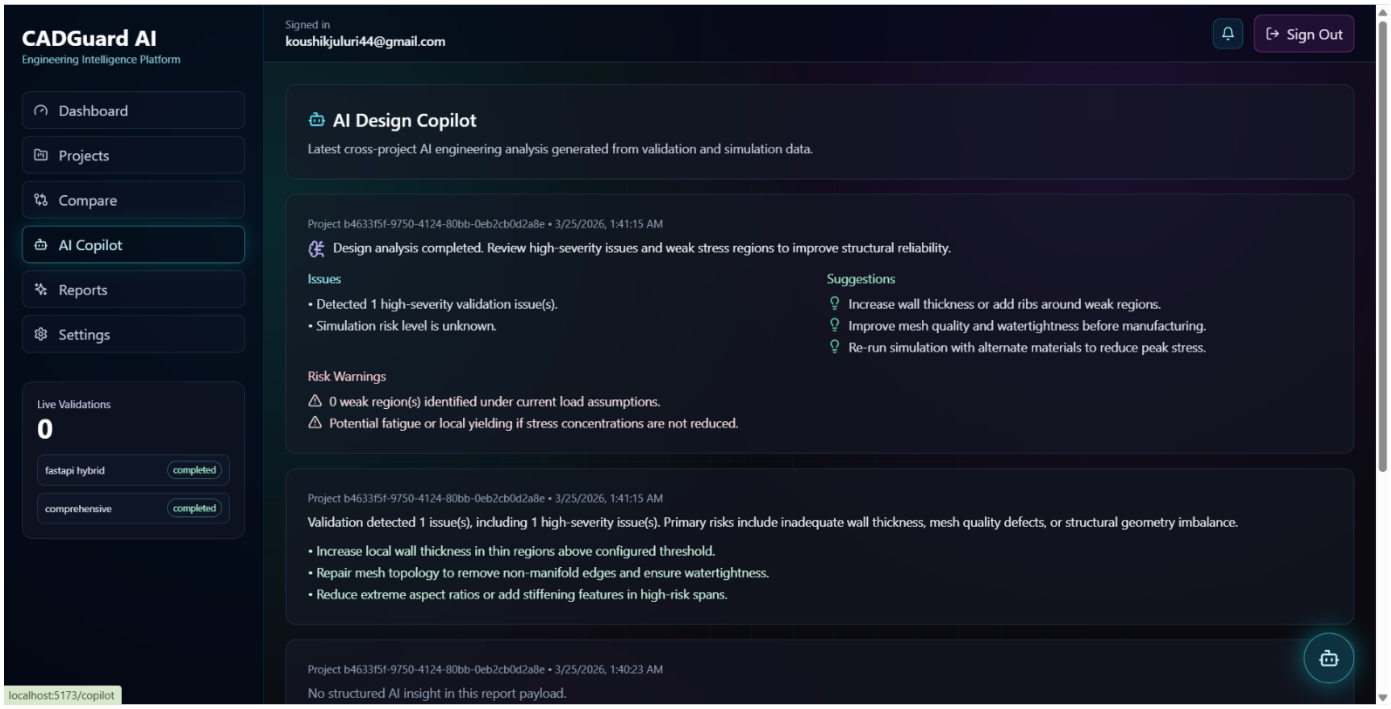
completed
Model Comparison
Upload two models to compare quality score, stress performance, and validation issues side-by-side.
Model A
Choose File
No file chosen
Model B
Choose File
No file chosen
Run Comparison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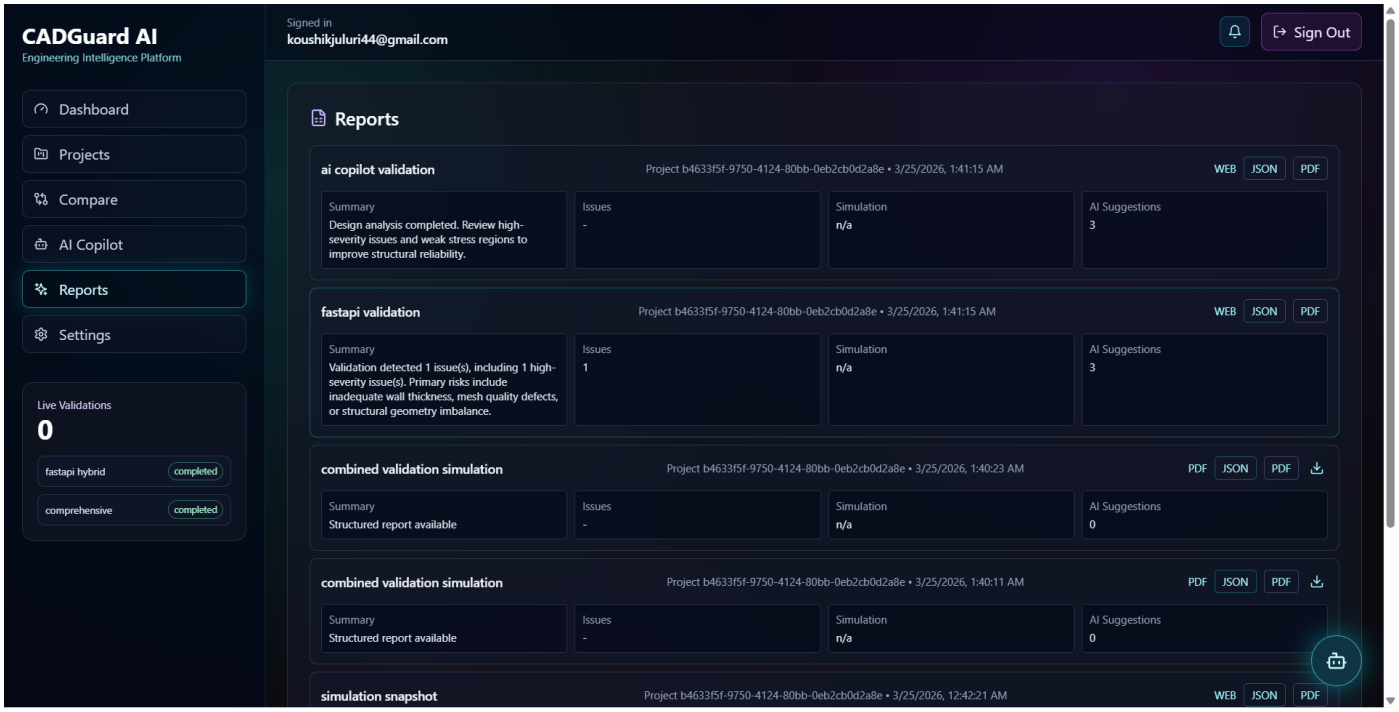
AI Design Copilot
Latest cross-project AI engineering analysis generated from validation and simulation data.
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:41:15 AM
Design analysis completed. Review high-severity issues and weak stress regions to improve structural reliability.
Issues
• Detected 1 high-severity validation issue(s).
• Simulation risk level is unknown.
Suggestions
Increase wall thickness or add ribs around weak regions.
Improve mesh quality and watertightness before manufacturing.
Re-run simulation with alternate materials to reduce peak stress.
Risk Warnings
0 weak region(s) identified under current load assumptions.
Potential fatigue or local yielding if stress concentrations are not reduced.
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:41:15 AM
Validation detected 1 issue(s), including 1 high-severity issue(s). Primary risks include inadequate wall thickness, mesh quality defects, or structural geometry imbalance.
• Increase local wall thickness in thin regions above configured threshold.
• Repair mesh topology to remove non-manifold edges and ensure watertightness.
• Reduce extreme aspect ratios or add stiffening features in high-risk spans.
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:40:23 AM
No structured AI insight in this report payload.
localhost:5173/copilot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Dashboard
Projects
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AI Copilot
Reports
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Live Validations
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fastapi hybrid
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Reports
ai copilot validation
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:41:15 AM
WEB JSON PDF
Summary
Design analysis completed. Review high-severity issues and weak stress regions to improve structural reliability.
Issues
-
Simulation
n/a
AI Suggestions
3
fastapi validation
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:41:15 AM
WEB JSON PDF
Summary
Validation detected 1 issue(s), including 1 high-severity issue(s). Primary risks include inadequate wall thickness, mesh quality defects, or structural geometry imbalance.
Issues
1
Simulation
n/a
AI Suggestions
3
combined validation simulation
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:40:23 AM
PDF JSON PDF
Summary
Structured report available
Issues
-
Simulation
n/a
AI Suggestions
0
combined validation simulation
Project b4633f5f-9750-4124-80bb-0eb2cb0d2a8e • 3/25/2026, 1:40:11 AM
PDF JSON PDF
Summary
Structured report available
Issues
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Simulation
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AI Suggestions
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simulation snapshot
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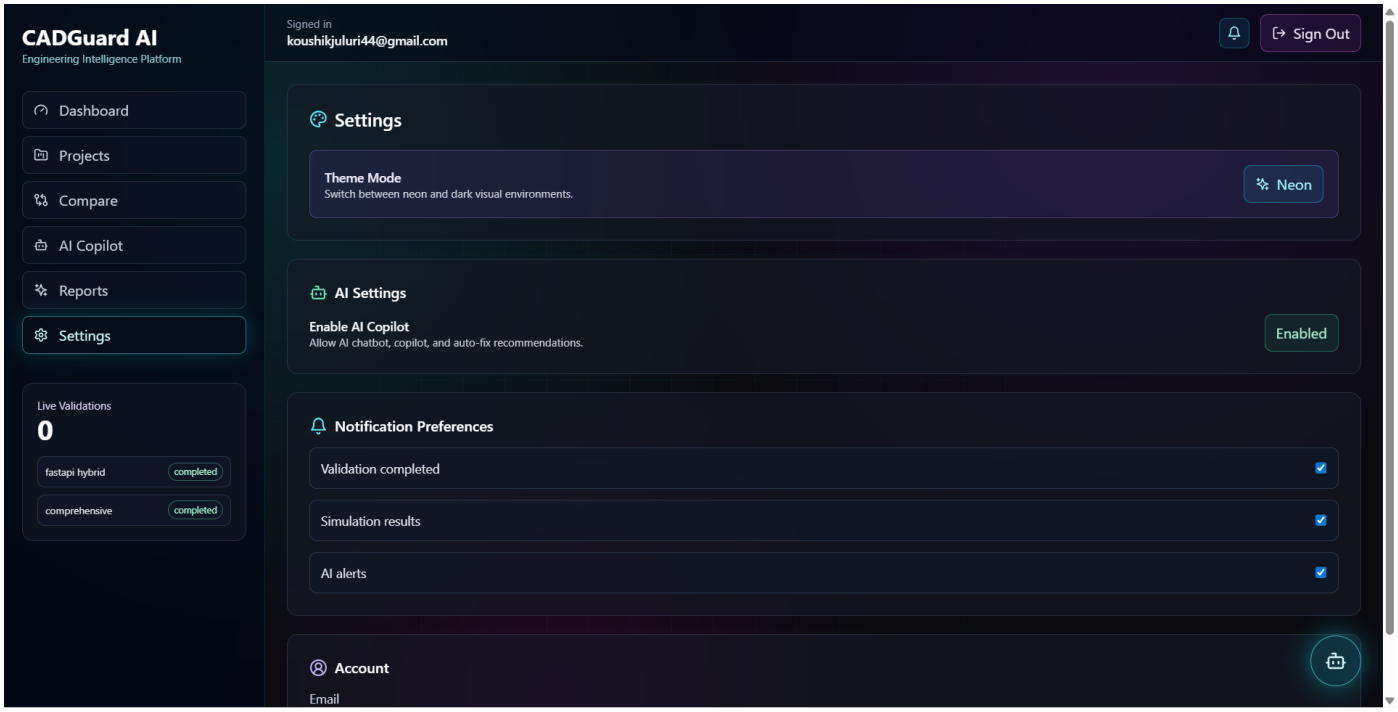
WEB JSON PDF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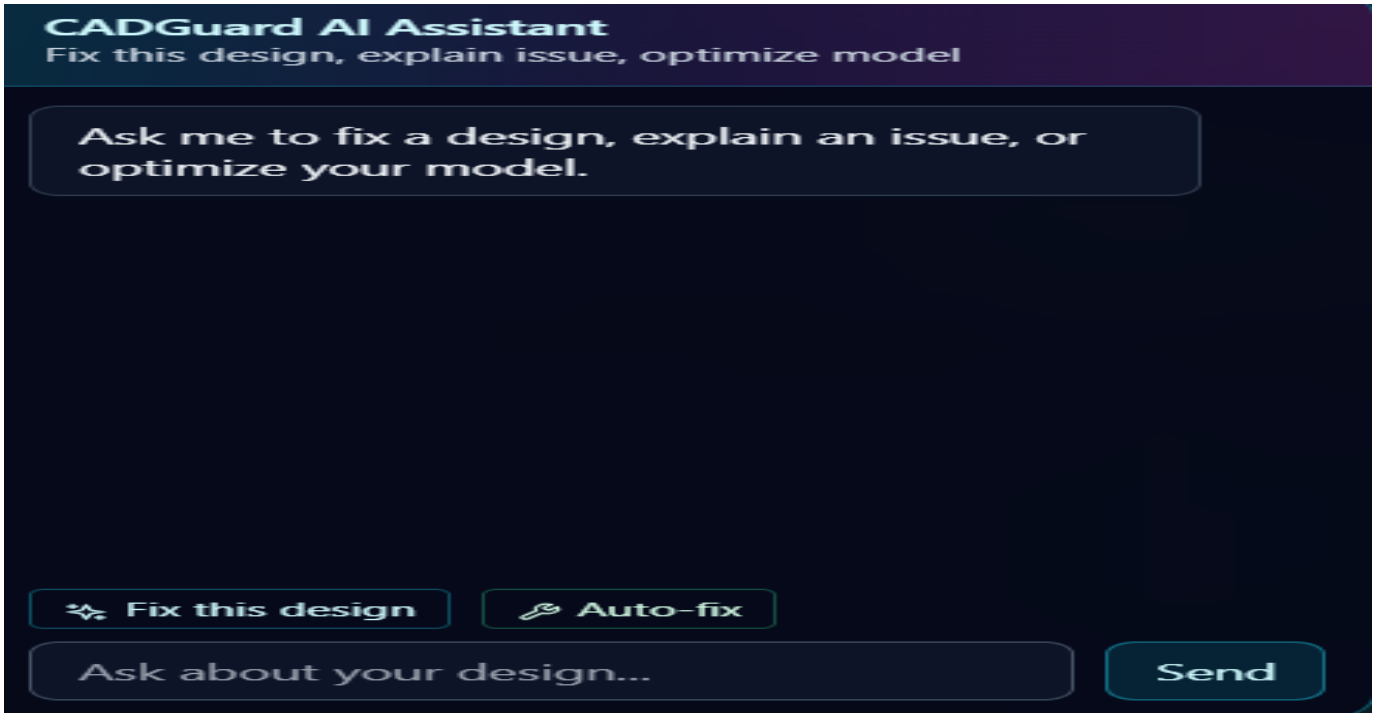
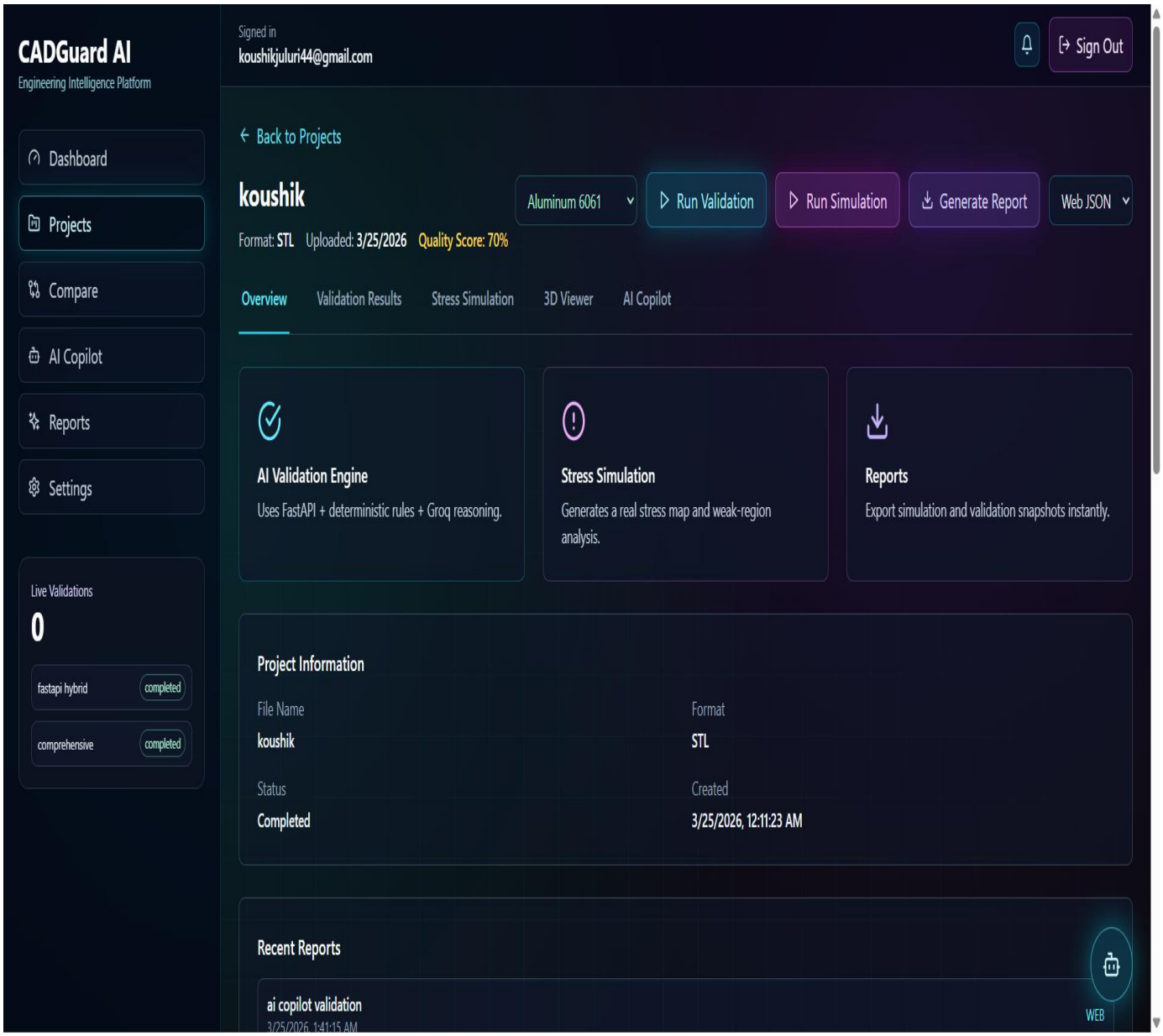
Sign Out
Settings
Theme Mode
Switch between neon and dark visual environments.
Neon
AI Settings
Enable AI Copilot
Allow AI chatbot, copilot, and auto-fix recommendations.
Enabled
Notification Preferences
Validation completed
Simulation results
AI alerts
Account
Email


CADGuard AI Assistant
Fix this design, explain issue, optimize model
Ask me to fix a design, explain an issue, or optimize your model.
Fix this design
Auto-fix
Ask about your design...
Send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Back to Projects
koushik
Aluminum 6061
Run Validation
Run Simulation
Generate Report
Web JSON
Format: STL Uploaded: 3/25/2026 Quality Score: 70%
Overview
Validation Results
Stress Simulation
3D Viewer
AI Copilot
AI Validation Engine
Uses FastAPI + deterministic rules + Groq reasoning.
Stress Simulation
Generates a real stress map and weak-region analysis.
Reports
Export simulation and validation snapshots instantly.
Project Information
File Name
koushik
Format
STL
Status
Completed
Created
3/25/2026, 12:11:23 AM
Recent Reports
ai copilot validation
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CHAPTER 4
CONCLUSION

CADGaurd represents a modern approach to engineering validation by bringing multiple critical processes into a single, unified platform. Instead of relying on separate tools for design checking, simulation, and analysis, it enables engineers to perform all these tasks within one seamless workflow. This integration not only saves time but also reduces errors caused by fragmented processes, making the overall design cycle more efficient and reliable.

The platform's use of deterministic geometry validation ensures that common manufacturability issues are identified early in the design phase. By detecting problems such as improper wall thickness, non-manifold edges, and structural inconsistencies, CADGaurd helps engineers prevent costly redesigns later. Early risk detection plays a crucial role in improving product quality and ensuring that designs meet practical production requirements.

In addition to rule-based validation, CADGaurd incorporates lightweight digital twin–style stress analysis to provide quick insights into structural performance. This allows users to identify weak regions and assess risk levels without the need for complex and time-consuming simulations. As a result, engineers can make faster, data-driven decisions while maintaining a balance between speed and analytical depth.

A key strength of CADGaurd lies in its AI-powered assistance, which transforms technical outputs into clear, understandable insights. By offering explanations, prioritized issues, and actionable recommendations, the platform reduces the cognitive effort required to interpret results. Features such as AI copilot guidance and chatbot interaction further enhance user experience, making advanced engineering analysis more accessible and intuitive.

Overall, CADGaurd stands out as a scalable, secure, and production-ready solution for modern engineering workflows. Its modular architecture, real-time capabilities, and continuous improvement loop support collaborative and iterative design processes. By enabling engineers to move from raw CAD input to informed decision-making efficiently, CADGaurd contributes to higher design quality, reduced development time, and improved innovation outcomes.